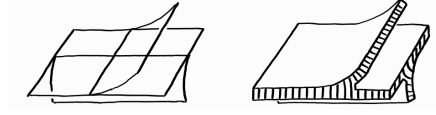


INTRODUCTION TO DYNAMIC PAIRS

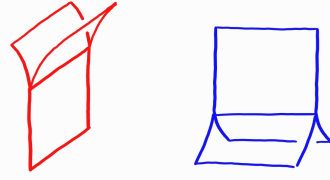
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9 MAR 2026

Recall that a **branched surface** on a closed orientable 3-manifold M is a 2-complex B smoothly modeled on a point in the following picture.



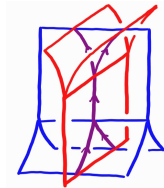
The **branch locus** of B is the set of non-manifold points. This naturally has the structure of a 4-valent graph. The **triple points** of B are the vertices of this graph. The **max coorientation** of each edge of $\text{brloc}(B)$ is the coorientation on B pointing from the side with more sectors to the side with less sectors.

A **stable dynamic branched surface** is a branched surface B^s for which there exists a C^0 vector field V on M that is tangent to B^s and negatively transverse to $\text{brloc}(B^s)$. An **unstable dynamic branched surface** is a branched surface B^u for which there exists a C^0 vector field V on M that is tangent to B^u and positively transverse to $\text{brloc}(B^u)$. A choice of V in this definition is called a **dynamic vector field** for B^s/u .



Morally, a **dynamic pair** is a pair (B^s, B^u) consisting of one stable dynamic branched surface and one unstable dynamic branched surface such that:

- (I) B^s and B^u meet transversely.
- (II) There is a common dynamic vector field to B^s and B^u .
- (III) Each complementary region of $B^s \cup B^u$ in M is simple.
- (IV) Each complementary region of $\tau = B^s \cap B^u$ in B^s and in B^u is simple.

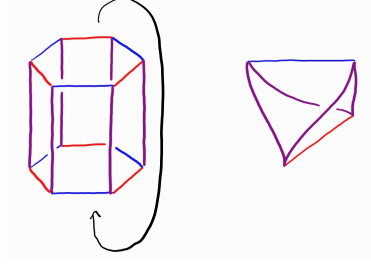


The complementary regions of $B^s \cup B^u$ are 3-manifolds with boundary divided into **s**- and **u**-faces. Furthermore, two **s**-faces meet along **ss**-cusp edges, two **u**-faces meet along **uu**-cusp edges, a **s**-face

and a **u**-face meet along **su**-corner edges. The **su**-corner edges are oriented by the dynamic vector field.

A **dynamic solid torus** is a solid torus whose boundary is divided into $2n \geq 4$ annulus faces, alternatingly labeled **s** and **u**, with the **su**-circles oriented coherently. A **pinched tetrahedron** is a tetrahedron with two faces labeled **s** and two faces labeled **u**, with the **su**-edges oriented from the **ss**-edge to the **uu**-edge.

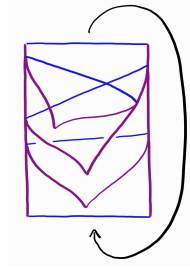
- (III) Each complementary region of $B^s \cup B^u$ in M is a dynamic solid torus or a pinched tetrahedron. Moreover, the dynamic solid tori do not share faces.



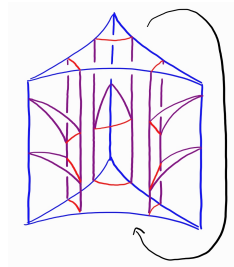
The intersection $\tau = B^s \cap B^u$ is an unstable train track on B^s . This means that its edges are oriented in a way such that converging switches lie away from $\text{brloc}(B^s)$ and diverging switches lie on $\text{brloc}(B^u)$. The complementary regions of τ in B^s are thus branched surfaces with boundary divided into converging train tracks joined along cusp points.

An **annulus with tongues** is a branched surface constructed by attaching 1-cusped triangles to an annulus along a coherent orientation. A **Möbius band with tongues** is defined similarly.

- (IV) Each complementary region of $\tau = B^s \cap B^u$ in B^s and in B^u is an annulus or Möbius band with tongues. In this case we say that τ **fills** B^s and B^u .

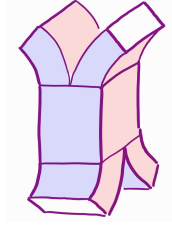


From the axioms of a dynamic pair, one can show that the complementary regions of B^s and of B^u are **cusped solid tori**. The components of $M \setminus (B^s \cup B^u)$ and $B^{s/u} \setminus \tau$ fit together within each cusped solid torus in the following way.



Theorem 1 (Mosher). *A closed orientable 3-manifold M admits a pseudo-Anosov flow ϕ if and only if it admits a dynamic pair (B^s, B^u) . Moreover, in this case, B^s carries the stable foliation of ϕ and B^u carries the unstable foliation of ϕ .*

Idea of $(\Leftarrow)$. Replace the edges of the dynamic train track $\tau = B^s \cap B^u$ by flow boxes. Glue up the vertical faces of the flow boxes by collapsing the annuli and Möbius bands with tongues. $\square$



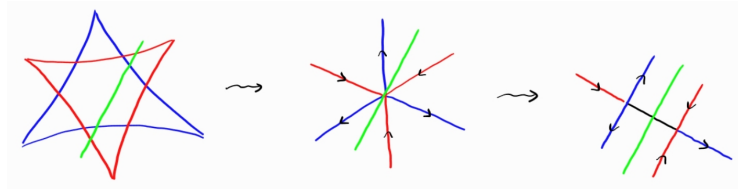
We say that a dynamic pair (B^s, B^u) is **vertical** with respect to a cooriented branched surface B^H if

- (i) there exists a dynamic vector field that is transverse to B^H , and
- (ii) the restriction of B^H to each complementary region of B^s and of B^u does not carry any peripheral annuli.

Theorem 1 can be upgraded as follows.

Theorem 2. *A closed orientable 3-manifold M admits a pseudo-Anosov flow ϕ almost transverse to a cooriented foliation $\mathcal{F}$ if and only if it admits a dynamic pair (B^s, B^u) that is vertical with respect to a cooriented branched surface B^H carrying $\mathcal{F}$.*

The ‘almost’ arises from the following local picture.



The main theorem is the following.

Theorem 3 (Mosher). *Let B^H be a branched surface carrying a sutured hierarchy $M = Q_0 \rightsquigarrow \dots \rightsquigarrow Q_{n+1} \cong S_{n+1} \times I$ of a closed orientable atoroidal 3-manifold M . Then, up to splitting B^H , M admits a dynamic pair (B^s, B^u) that is vertical with respect to B^H .*

Corollary 4 (Gabai, Mosher). *Let $\mathcal{F}$ be a finite depth foliation on a closed orientable atoroidal 3-manifold M . Then M admits a pseudo-Anosov flow almost transverse to $\mathcal{F}$.*

We explain how to extend the definition of dynamic pairs to sutured manifolds.

The definition of stable and unstable dynamic branched surfaces is the same, with the requirement that we restrict to vector fields that are positively transverse to $R_{\pm}$ and tangent to γ .

The complementary regions of $B^s \cup B^u$ are now divided into **s**-, **u**-, **p**-, **m**-, and **b**-faces, where the **p**-/ **m**-/ **b**-faces are the remains of $R_+/R_-/\gamma$ respectively.

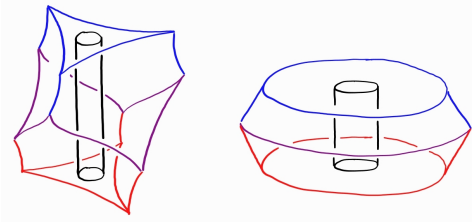
A **dynamic torus shell** is a thickened torus with one boundary component divided into $2n \geq 2$ annulus faces, alternatingly labeled **s** and **u**, and the other boundary component labeled **b**.

A **drum** with base S is $S \times I$ with $S \times \{1\}$ labeled **p**, $S \times \{0\}$ labeled **m**, and each component of $\partial S \times I$ divided into faces in one of the following ways:

- Alternating one-cusped pentagon **s**- and **u**-faces.
- One annulus **s**- and one annulus **u**-face.
- One **b**-face.

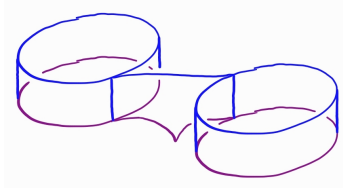
such that $\text{ind}(\partial_{\mathbf{p}}S) = \text{ind}(\partial_{\mathbf{m}}S) \leq 0$.

- (III) Each complementary region of $B^s \cup B^u$ in M is a dynamic solid torus, a dynamic torus shell, a pinched tetrahedron, or a drum. Moreover, the dynamic solid tori and torus shells do not share faces.

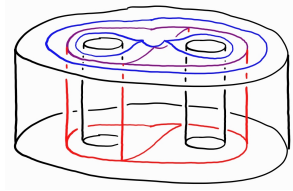


A component of $B^s \setminus \tau$ is **transient** if it is of the form $\beta \times I$ with $\beta \times \{0\}$ lying along R_- and $\beta \times \{1\}$ lying along τ . Note that β has a natural structure of a spiraling train track, i.e. β consists of a collection of circles $\text{circ}(\beta)$ and a union of branches that spiral into the circles in a coherent orientation.

- (IV) Each complementary region of $\tau = B^s \cap B^u$ in B^s and in B^u is an annulus or Möbius band with tongues, or transient. In this case we say that τ fills B^s and B^u .



From the axioms of a dynamic pair, one can show that the complementary regions of B^s and of B^u are **cusped solid tori** and **cusped products**. The components of $M \setminus (B^s \cup B^u)$ and $B^{s/u} \setminus \tau$ fit together within each cusped product in the following way.



The proof of [Theorem 3](#) is by induction.

Theorem 5 (Mosher, Landry-T.). *Let $Q_n \rightsquigarrow Q_{n+1} \cong S_{n+1} \times I$ be a sutured decomposition. Suppose Q_n is atoroidal. Then Q_n admits a dynamic pair (B_n^s, B_n^u) that is vertical with respect to a branched surface carrying the sutured decomposition.*

Theorem 6 (Mosher). *Let $Q_{k-1} \rightsquigarrow Q_k$ be a sutured decomposition. Suppose Q_k admits a dynamic pair (B_k^s, B_k^u) that is vertical with respect to a branched surface B_k^H and suppose Q_{k-1} is atoroidal. Then Q_{k-1} admits a dynamic pair (B_{k-1}^s, B_{k-1}^u) that is vertical with respect to a branched surface B_{k-1}^H carrying a splitting of B_k^H and the sutured decomposition.*